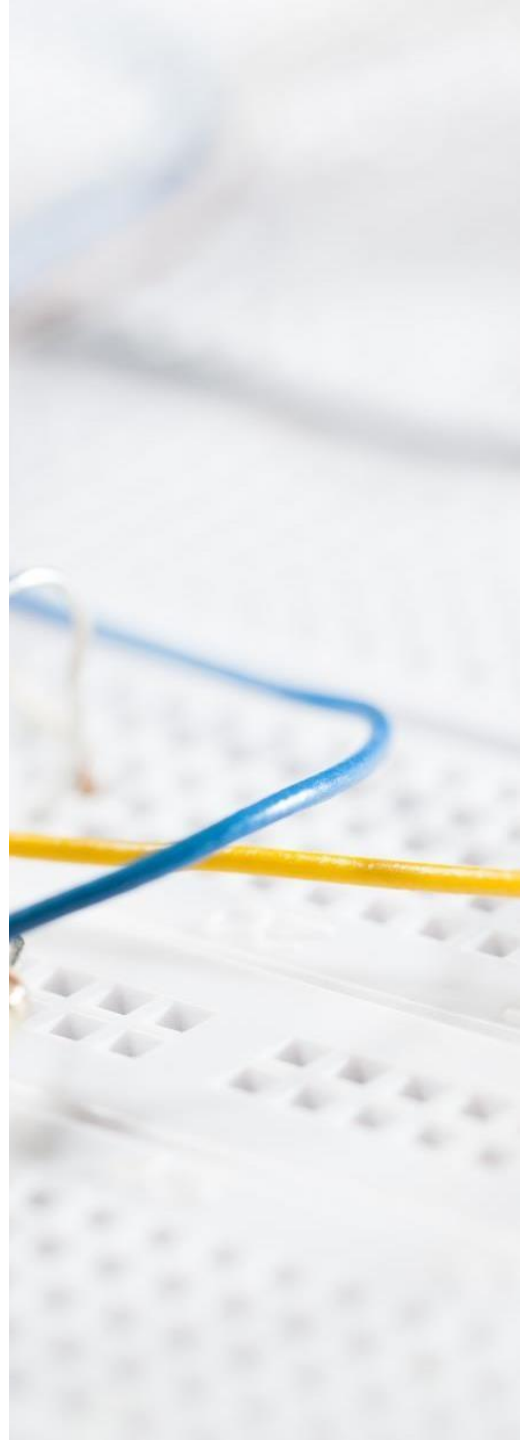
Introduction to Computers System

Section.2.



Agenda

- Number Systems
- Types of Number Systems
- Binary Number System
- Decimal Number System
- Conversion Rules of Number Systems



Number Systems

Number Systems

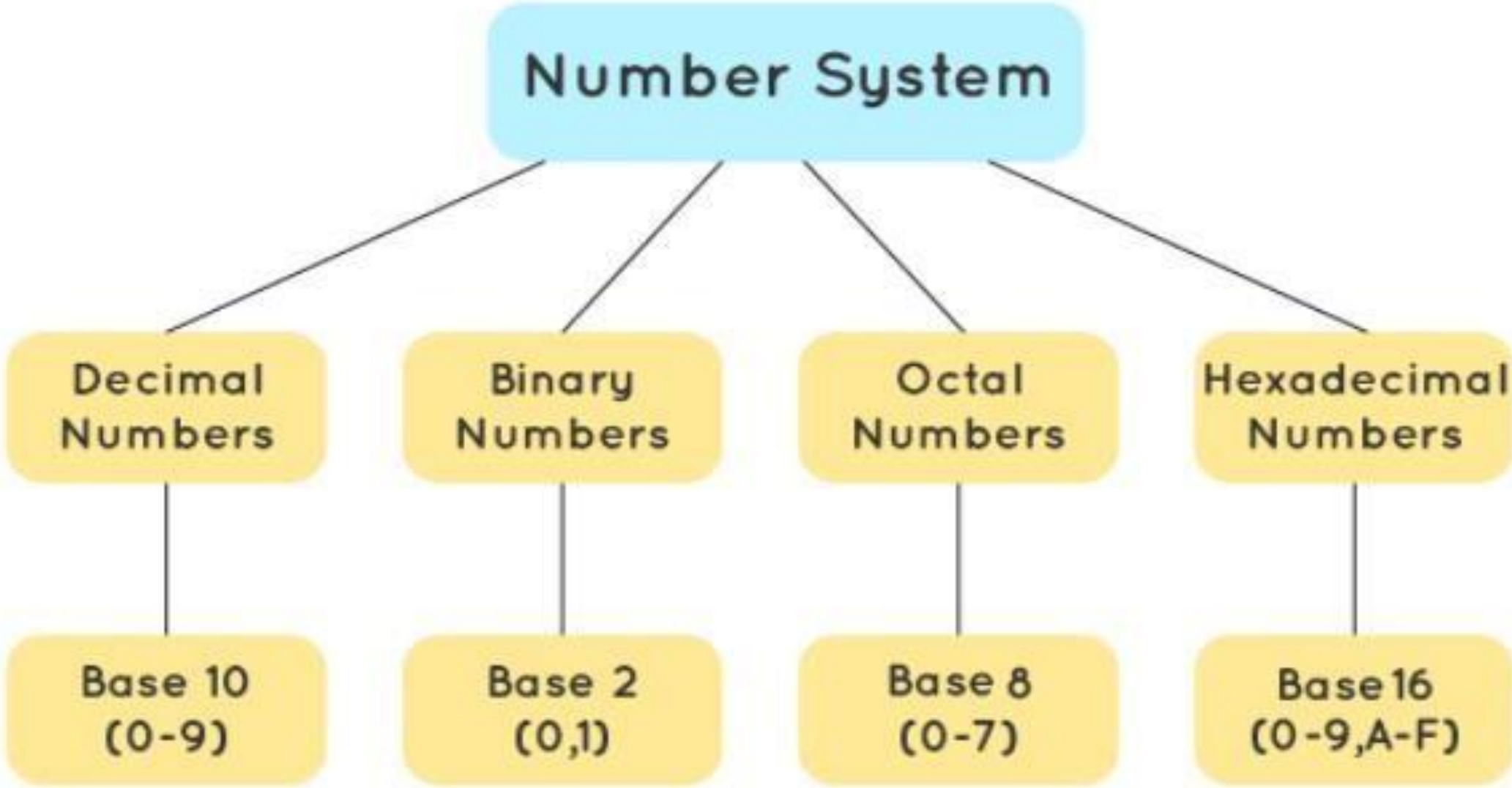
- A number system is a system representing numbers.
- A number is a mathematical value used for counting and measuring objects, and for performing arithmetic calculations.

Types of Number Systems

Types of Number Systems

- There are different types of number systems in which the four main types are:
    - Binary number system.
    - Octal number system.
    - Decimal number system.
    - Hexadecimal number system.

- We will study each of these systems one by one in detail.


Number System
Decimal Numbers
Binary Numbers
Octal Numbers
Hexadecimal Numbers
Base 10 (0-9)
Base 2 (0,1)
Base 8 (0-7)
Base 16 (0-9,A-F)

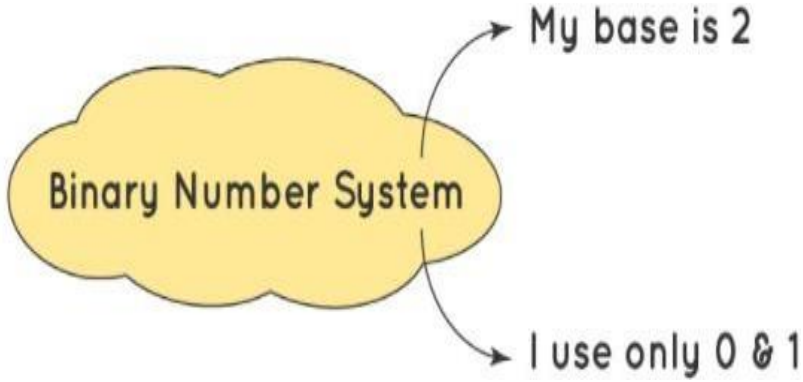
Binary Number System

Binary Number System:

- The binary number system uses only two digits: 0 and 1.
- The numbers in this system have a base of 2.
- it does not deal with other numbers such as 2,3,4,5 and so on.

- <u>For example:</u>

  10001, 111101, 1010101 are a binary number.



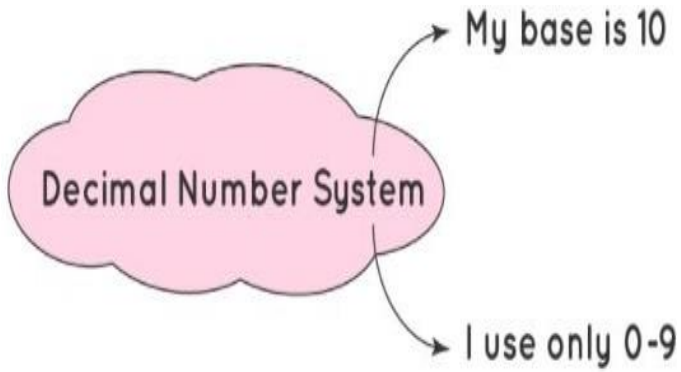
Decimal Number System

Decimal Number System

- The decimal number system has a base 10 because it uses ten digits from 0 to 9.

- **For example:**

  $(723)_{10}$, $(32)_{10}$, $(4257)_{10}$ are decimal number.



Name	Base	Symbols	Example
Decimal	10	0,1,2,3,4,5,6,7,8,9	$(2795)_{10}$
Binary	2	0,1	111000010
Octal	8	0,1,2,3,4,5,6,7	$(1576)_{8}$
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F	3DB

Conversion Rules of Number Systems

Binary System Conversions

- **Binary to Decimal:**

Example: **1101001**

Determine the weight of each bit that is a 1, and then sum the weights to get the decimal fraction.

Weights	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1

$2^6 = 64$	$2^5 = 32$	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
1	1	0	1	0	0	1

Now, adding all these:

=(1*1) + (2*0) + (4*0) + (8*1) + (16*0) + (32*1) + (64*1)

= 1 + 0 + 0 + 8 + 0 + 32 + 64 = $(105)_{10}$

Convert the binary number 10010001 to decimal.

Binary weights.

Positive Powers of Two (Whole Numbers)									Negative Powers of Two (Fractional Number)					
2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}	2^{-6}
256	128	64	32	16	8	4	2	1	1/2 0.5	1/4 0.25	1/8 0.125	1/16 0.625	1/32 0.03125	1/64 0.015625

Example: 0.1011

$$\begin{aligned} \text{Weight:} \quad & 2^{-1} \quad 2^{-2} \quad 2^{-3} \quad 2^{-4} \\ \text{Binary number:} \quad & 0.1 \quad 0 \quad 1 \quad 1 \\ 0.1011 &= 2^{-1} + 2^{-3} + 2^{-4} \\ &= 0.5 + 0.125 + 0.0625 = \mathbf{0.6875} \end{aligned}$$

- Convert 223_{10} *into binary system.*

A. <u>11011111</u>

B. 11101111

C. 11011110

- **Binary to Octal:**

Example: **110101**

Octal means 8

Staring from the least significant bi ,

we will **make a group of 3 bits:**

Second group	First pair
110	101
6	5

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

so, the equivalent octal number system will be $(65)_8$.

Convert the binary number 1010101000111110010 to octal.

- **Binary to Hexadecimal:**

Example: 101010001

We know that hexadecimal means 16.

we will start from the least significant bit and **make groups of 4 bits**:

Third pair	Second pair	First pair
0001	0101	0001
1	5	1

So, the hexadecimal equivalent will be $(151)_{16}$.

Example: 110101101101

Haxa	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Decimal System Conversions

- **Decimal to Binary:**

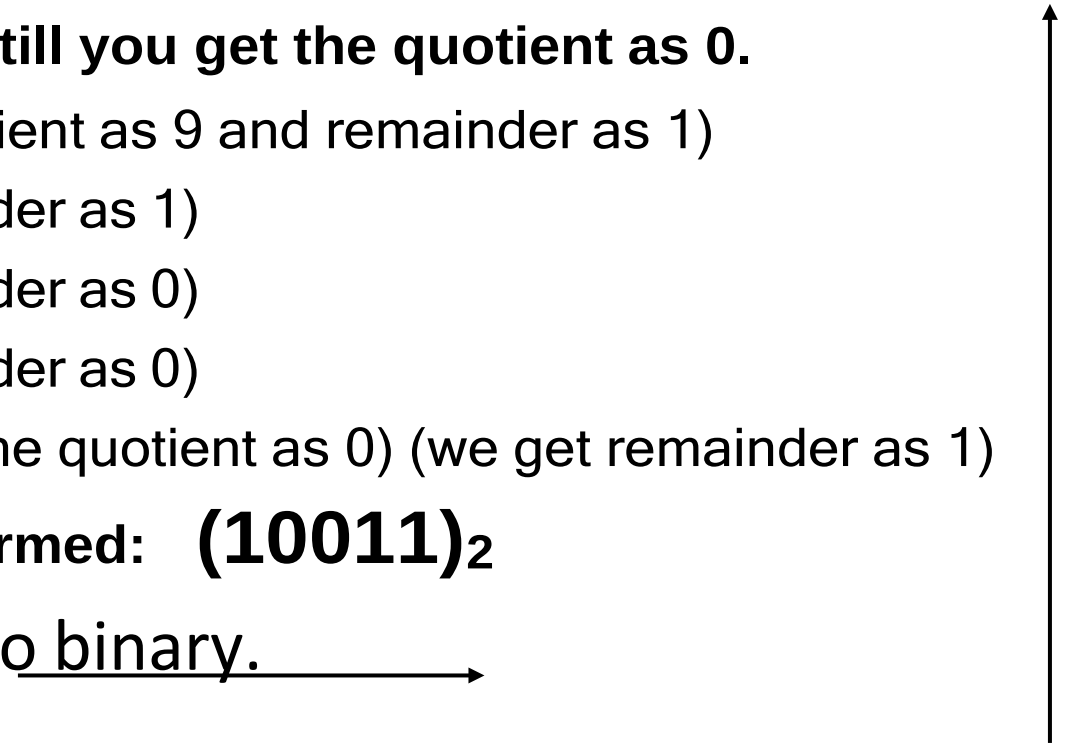
  Example: $(19)_{10}$

  Divide 131 by 2 and continue till you get the quotient as 0.

  - Divide 19 by 2 (we get the quotient as 9 and remainder as 1)
  - Divide 9 by 2 (we get 4, remainder as 1)
  - Divide 4 by 2 (we get 2, remainder as 0)
  - Divide 2 by 2 (we get 1, remainder as 0)
  - and finally Divide 1 by 2 (with the quotient as 0) (we get remainder as 1)

  write the binary system so formed: $(10011)_2$

Convert decimal number 39 to binary.



- **Decimal to Octal Conversion:**

  Example: **$(359)_{10}$**

  Divide 359 by 8 till you get the quotient as 0.

  - Dividing 359 by 8 (we get the quotient as 44.875 (0.875* 8) =7, so remainder as 7)
  - Divide 44 by 8 (we get the quotient as 5.5 (0.5 * 8)= 4, so remainder as 4)
  - Divide 5 by 8 (we get the quotient as 0.625 (0.625 *8)= 5, we get the remainder as 5)

So, the octal equivalent of $(359)_{10}$ is $(547)_8$.

Example: $(243)_{10}$, $(87)_{10}$

		Remainder
8	87	7
8	10	2
8	1	1
		(127)

- **Decimal to Hexadecimal:**

  Example: **$(650)_{10}$**

  - Divide 650 by 16, we get the quotient as 40.625 (0.625*16) =10 , so remainder as 10 or A.
  - Divide 40 by 16, we get the quotient as 2.5 (0.5 *16) =8 ,so remainder as 8.
  - Divide 2 by taking the quotient as 0.125 (0.125*16)= 2, so remainder as 2.

  So, the hexadecimal equivalent of $(650)_{10}$ is $(28A)_{16}$.

Example: $(243)_{10}$

		Remainder
16	243	3
16	15	15
		(F3)

Octal System Conversions

- **Octal to Decimal:**

Example: $(247)_8$

starting from right to left .

$8^2 = 64$	$8^1 = 8$	$8^0 = 1$
2	4	7

= (7*1) + (4*8) + (2*64)

= 7 + 32 + 128 = $(167)_{10}$

Example:$(127)_8$

- **Octal to Binary:**

Example: **$(247)_8$**

2	4	7
010	100	111

$(010100111)_2$

The zero on the left has no value, so

$(10100111)_2$

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example: **$(375)_8$**

- **Octal to Hexadecimal:**

Steps:

1. Convert from Octal to Binary.
2. Convert from Binary to Hexadecimal.

Example: $(753)_8$

Step1:

7	5	3
111	101	011
$(111101011)_2$		

Step2:

$(1\ 1110\ 1011)_2 \rightarrow (?)_{16}$		
0001	1110	1011
1	E	B
$(1EB)_{16}$		

Octal	Binary	Decimal	Haxa	Binary
0	000			
1	001	1	1	0001
2	010			
3	011	3	3	0011
4	100			
5	101	5	5	0101
6	110			
7	111	7	7	0111
		9	9	1001
		10	A	1010
		11	B	1011
		12	C	1100
		13	D	1101
		14	E	1110
		15	F	1111

Hexadecimal System Conversions

- **Hexadecimal to Decimal:**

Example: $(3A)_{16}$

	$16^1 = 16$	$16^0 = 1$
6)	3=3	A=10

$= (10*1) + (3*$

$= 10 + 48 = (58)_8$

Decimal	Haxa
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

- **Hexadecimal to Binary:**

Example: $(5B)_{16}$

5 = 5	B= 11
0101	1011
$(01011011)_2$	

Decimal	Haxa	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

- **Hexadecimal to Octal:**

Steps:

1. Convert from Hexa to Binary.
2. Convert from Binary to Octal.

Example: $(B2E)_{16}$

Step1:

B = 11	2 = 2	E = 14
1011	0010	1110
$(101100101110)_2$		

Step2:

$(101\ 100\ 101\ 110)_2$			
101	100	101	110
5	4	5	6
$(5456)_8$			

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Decimal	Haxa	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Question

- Convert the binary number 00101101 to decimal.
- Convert the binary number 10011110111110011100 to hexadecimal.

45

4F79C

- Convert the binary number 1010101000111110010 to octal.
- Convert decimal 453 to binary.
- Convert decimal 10 to octal.
- Convert decimal 591 to hexadecimal.

1250762

111000101

12

24F

- The octal number system is a weighted system with ten digits. FALSE
- The binary number system is a weighted system with two digits. TRUE
- In hexadecimal, 9 +1 = 10. FALSE
- The hexadecimal number system has 16 characters, five of which are alphabetic characters. FALSE

- The binary number 1000110101000111011111 can be written in hexadecimal as

(a) AD467 **(b)** 8C46F **(c)** 8D46F **(d)** AE46F

Assignment

1. Convert 300_{10} into the binary system.
2. Convert $5BC_{16}$ into the decimal system.
3. Convert 144_{8} into the hexadecimal system.

Due Data: 1-11-2024 at 11:59 p.m. (Delivered Online)



Thank you

